



NETFLIX PRIZE DATASET

Building a Personalized Movie Recommendation System

Comparing neighbourhood-based and latent-factor models on 1.2M ratings — evaluated for both rating accuracy (RMSE) and ranking quality (MAP@10).

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1.2M

ratings

2,634

users

1,287

movies

2

models compared

PROBLEM OVERVIEW

Two goals, not one

A recommender must do two related but distinct jobs. A model can be strong at one and weak at the other — so both must be measured.



Rating Prediction

Estimate the score a user would give an unseen movie.

Measured by RMSE



Recommendation Ranking

Place the movies a user will truly like at the top of a 10-item list.

Measured by MAP@10



The dataset & the task

Netflix Prize data (24M+ ratings in file 1). A movie counts as **relevant** if the user rated it ≥ 3.5 . We learn from each user's past and predict their future.

Data pipeline & two contrasting models



STEP 1

Smart Subsetting

Filtered to active users (≥ 400 ratings) & popular movies (≥ 2000). 24M $\rightarrow$ 1.2M ratings, far denser.



STEP 2

Time-based Split

Per-user 80/20 by date: train on the past, test on the most recent 20%. No data leakage.



STEP 3

Two Models

A neighbourhood method vs a latent-factor method, evaluated identically for a fair comparison.



Item-Based CF

NEIGHBOURHOOD

- Movies similar if rated alike (cosine similarity)
- Predict from 30 nearest rated movies
- Simple & highly interpretable



SVD

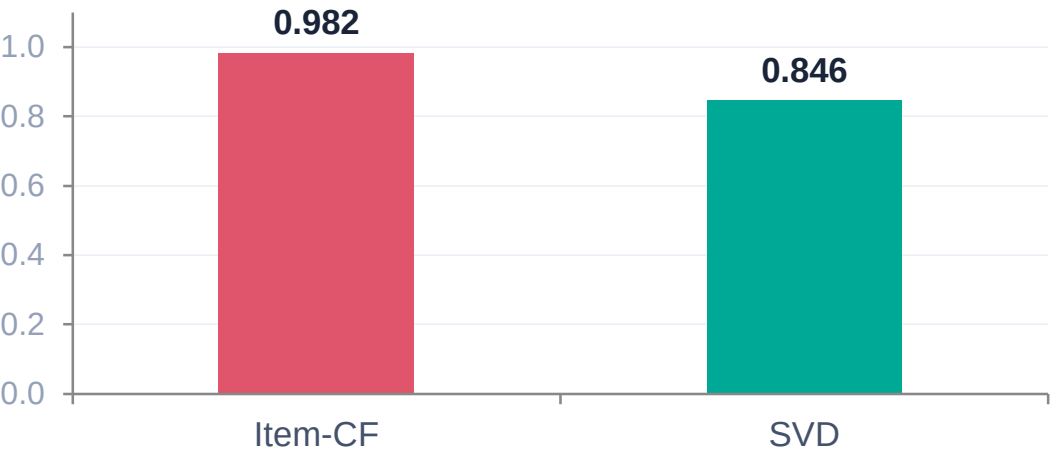
LATENT FACTOR

- Learns hidden taste factors per user & movie
- Captures global patterns; built with surprise
- Better-calibrated predictions

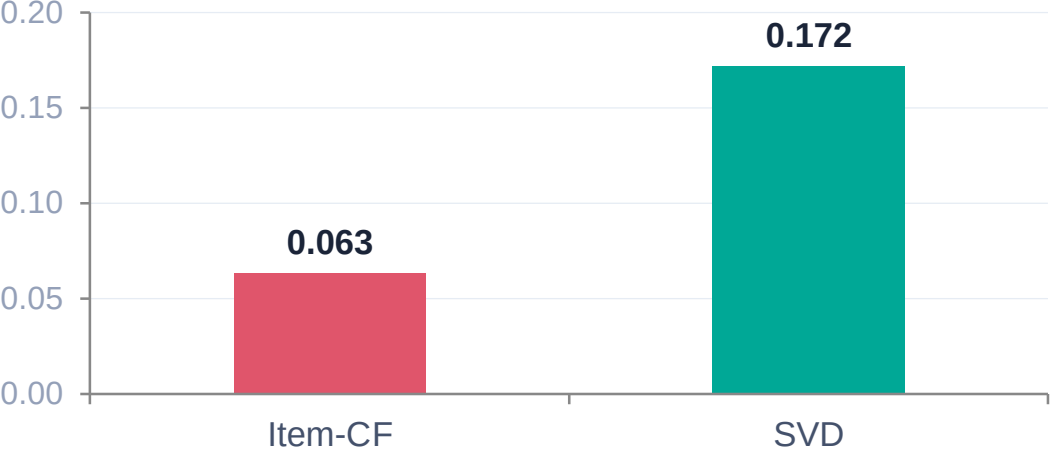
EXPERIMENTAL RESULTS

SVD wins on quality; Item-CF wins on cost

RMSE · lower is better



MAP@10 · higher is better



Quality winner: SVD

~2.7× higher MAP@10 and a lower RMSE — more accurate predictions and far better ranking.

Computational cost

Metric	Item-CF	SVD
Training time	~0	25.3 s
Predict / user	5.55 ms	7.25 ms
Memory	7.7 MB	27.8 MB

Item-CF is cheaper here, but its memory grows quadratically with catalogue size — SVD scales better.

What the models actually recommend

Across 200 users — average genuine “hits” in each Top-10:

1.44

Item-CF hits / user

62% reach ≥ 1 favourite

2.91

SVD hits / user

91.5% reach ≥ 1 favourite



Success — User 57186 (*prestige-film fan*)

SVD nailed 8 of 10 real favourites (LOTR, Batman Begins, Ray).
Item-CF matched none.



Failure — User 188190 (*very broad taste*)

50+ favourites across many genres — no 10-item list can cover them. Both models missed.

What the examples reveal

Item-CF weakness

Recommends the same niche arthouse cluster to very different users — weak personalization.

SVD strength

Recommendations clearly reflect each user’s individual taste profile.

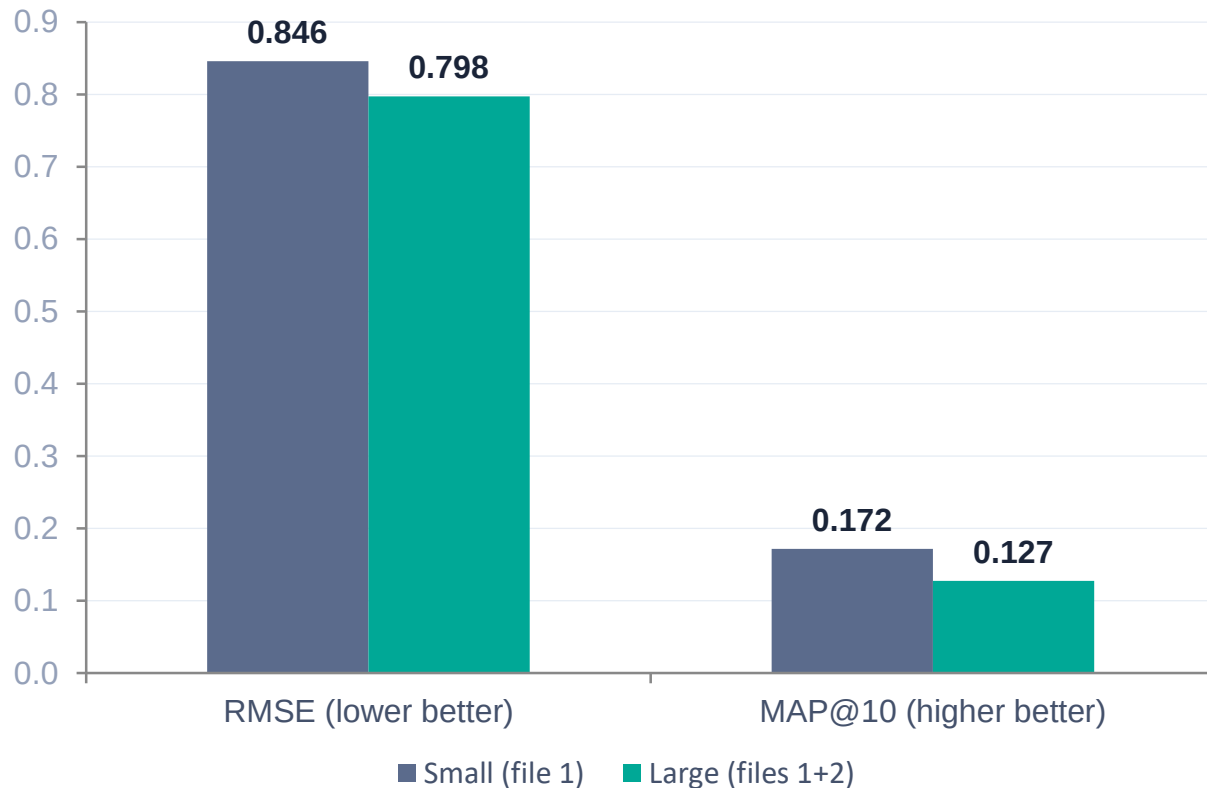
SVD limit

Struggles when a user’s tastes are too broad or inconsistent to summarise.

KEY INSIGHTS

More data isn't automatically better

Scale-up: file 1 → files 1+2 (24M → 51M ratings)



RMSE improved (0.85 → 0.80)

More ratings give SVD more signal — predictions get more accurate.



MAP@10 dropped (0.17 → 0.13)

Catalogue more than doubled (1,287 → 2,721) & sparsity rose — ranking is intrinsically harder.



The lesson

A single metric can mislead. Ranking difficulty depends on the candidate set, not just the model.

CONCLUSION

Which model, and when



Choose SVD

Best for recommendation quality

- Highest MAP@10 — ranks favourites near the top
- Well-calibrated, strongly personalized
- Scales better to large catalogues



Choose Item-CF

Best for interpretability & speed

- Every pick traces to similar movies you liked
- No training phase; fast to deploy
- Solid, transparent baseline



Future work

- Cold-start handling for new users & rare movies
- Richer metrics — NDCG, coverage, diversity
- Hybrid model: SVD accuracy + Item-CF explainability
- Scale to all four files with efficient training (ALS)

Summary

SVD delivers the best recommendations; Item-CF stays the interpretable baseline.

0.8455

SVD RMSE

vs 0.98 Item-CF

0.1717

SVD MAP@10

~2.7× Item-CF

91.5%

users reached

with ≥ 1 real favourite